

PROF. DR.-ING. MARK SCHUTERA

OWNING THE MIND

YOUR GUIDE TO SOVEREIGN AI AT HOME

UNFINISHED LECTURE NOTES

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These notes are, by their very nature, unfinished, and they improve with every reader. If you spot an error, disagree with a framing, or want to add a source, a question, or a position, open a pull request at https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes_sovereignai. Every contribution is welcome.



2026-08-20 · cheerful guava Habicht

"THERE AIN'T NO SUCH THING AS A FREE LUNCH."

THE MOON IS A HARSH MISTRESS, ROBERT A. HEINLEIN

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Introduction

COMPUTE MOVED OFF YOUR DESK. For most of computing history a program ran on hardware you owned. First the data moved into the cloud. Now the models run there too, on a meter, and what you send is retained unless you have contracted otherwise. Both things that you should own, your data and your compute, are now liabilities, and you have to pay for the privilege while relying on someone else's good faith.

SOVEREIGNTY IS NOT AUTARKY. National AI plans do not propose building the whole stack on their home turf. They name dependencies and the levers held against them¹. Europe holds the lithography that prints the world's chips, the United States most of the compute and the leading, frontier model weights. Most European plans though fund domestic compute to keep a lever of their own².

A HOUSEHOLD IS THE SAME SHAPE. You depend on the grid, a line, a fab in Taiwan, and weights someone else trained. What changes is which decisions are yours once you own the infrastructure the model runs on. This book is about building a household-scale private AI stack, from the metal up, which changes the dependencies to:

- cost paid once, then metered in kilowatt hours, not tokens;
- inference and data residency on machines you own;
- acquired technical knowledge and skills;
- weights on your own disk, and owning of the design space of your pipeline.

YOUR OWN MACHINE IS THE LEVER YOU CAN BUILD TODAY. All you need are some parts, this manual, and some tinkering. From then on the model runs where you put it, on data that stays where you put it. You pick the model, the quantisation, the runtime, and the moment to change any of them.

¹ D. Juijn et al. Europe 2031. Arq Foundation, 2026

² J. Prins and M. A. Bakker. Nederland: Nationaal AI deltaplan. Dutch Ministry of Economic Affairs, 2025

THIS BUILD IS FOR INFERENCE. AI inference loads model weights once and then streams tokens, so a single PCIe 3.0 lane keeps up. Training, fine tuning and image or video generation move tensors across the bus for the whole run. When facing the latter use-case you should target a server platform such as an AMD EPYC board, which gives six cards a full x16 link each.

The Rig

2026-08-20 · jazzy dragonfruit Hermelin

OWNING THE METAL.

THIS CHAPTER BUILDS ONE MACHINE: An open frame rig with a single RTX 3090 and its 24 GB, which is enough to serve a capable model for private usage. The rest of the setup is designed in a way so it is able to grow with your needs. Order the parts in Table 1 and fit them in the order of the sections that follow, until the last cable is seated and the machine is ready to be switched on. Day one is a few hours of plugging.

THE PARTS LIST is the Ryzen configuration of a proven quad 3090 local AI home server, built by me in 2026.

Component	Part	Pieces
Frame	open GPU rack frame	1
Motherboard	Gigabyte B650 Eagle AX	1
CPU	AMD Ryzen 5 9600X, AM5	1
CPU cooler	AM5 tower air cooler	1
Memory	G.SKILL Ripjaws S5 64 GB DDR5-6000	1
Storage	NVMe Gen4 x4 2 TB, M.2 2280	1
GPU	RTX 3090 24 GB, used	1 (4)
Risers, x1	PCIe 3.0 x1 riser, cards 2 to 4	0 (3)
Power supply	ASRock PG-1600G, 1600 W	1

Table 1: The bill of materials, in the order the parts go together. A bracketed number is what this board, frame, and supply will carry later.

The Motherboard

A GIGABYTE B650 EAGLE AX, a mainstream board that support our design decisions ³:

- To support the processor we will be choosing, the socket must be AM5.
- For the GPU the top x16 slot is best when running the sixteen PCIe lanes, short for peripheral component interconnect express, straight from the processor rather than through the chipset. A model's whole weight file crosses that link every time it loads, so model swapping will benefit from this wiring.
- Four slots have to be x16 sized, so we can grow the server in terms of GPU later on. Total VRAM, the video memory on the cards themselves, is the ceiling on model size, and we can extend that by up to 4 GPUs with this characteristics.
- For storage we need one M.2 socket wired PCIe 4.0 x4 or better straight from the processor, and this board has three, so a second drive later costs nothing today. That one drive carries everything the server accumulates, model weights, container volumes, embeddings, documents, and everything else data.

³ GIGABYTE. B650 EAGLE AX (rev. 1.2) motherboard specifications. Product datasheet, 2026

A SLOW SLOT therefore costs seconds per model swap, while this does not touch inference, we still care.

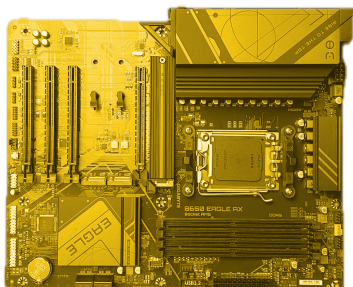


Figure 1: Gigabyte B650 Eagle AX

WORK ON IT FLAT. Lay the board on its own box, on the antistatic bag it shipped in, and populate it there: Processor, cooler, memory, and the drive all go in. This manual guides you step by step.

READ THE MANUALS. Every part ships one, and it holds the pin maps, socket order, and slot wiring that no build guide reproduces correctly.

The Processor and Cooler

AN AMD RYZEN 5 9600X, paired with an AM5 tower air cooler, a modest chip that supports our design decisions⁴.

- For inference we need very little compute on the CPU. Six Zen 5 cores, twelve threads, 3.9 GHz base and up to 5.4 boost, 32 MB of L3, and a 65 watt default thermal design power, is perfect for this. The GPUs carry the real work while the processor only feeds them and runs the container workloads layered on top.
- For the top slot to run a real x16, PCIe has to come from the processor itself rather than through the chipset.
- For the display we want integrated graphics on the chip itself, so the monitor never has to occupy a slot, and a 3090 stays free for the inference and AI model load.
- For the cooler we need an AM5 tower air cooler. The fan on the front of the fin stack sits over the first memory slot, so a fan with adjustable positioning is smart.

⁴ AMD. AMD Ryzen 5 9600X processor specifications. Product datasheet, 2026

WHY A TOWER AND NOT A LOW PROFILE BLOCK. An open frame has no lid, so the height a tower needs costs nothing here, and there is no case airflow for a flat block to borrow either. And towers are also more efficient and quieter than flat blocks.

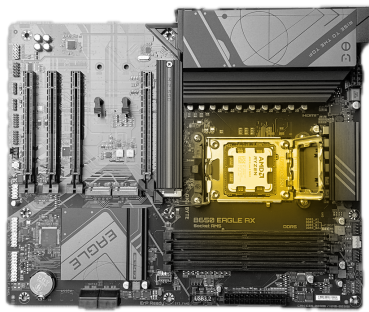


Figure 2: AMD Ryzen 5 9600X, AM5

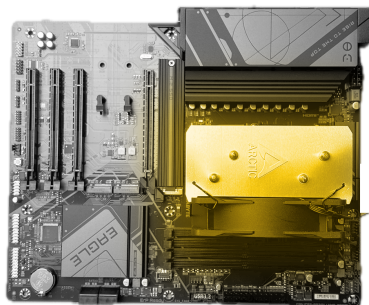


Figure 3: AM5 tower air cooler

SEAT IT by matching the small triangle on the chip's corner to the triangle on the socket, lower it flat without pressure, close the retention arm (Figure 2).

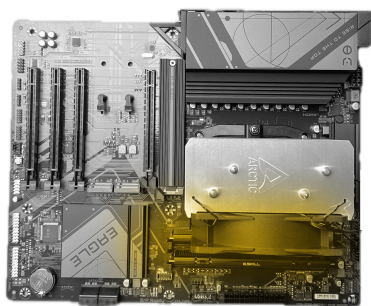
THERMAL PASTE is often pre-applied to the cooler, but if it is not, apply a small pea sized drop to the center of the chip, and let the cooler spread it when you mount it. Too much paste behaves like a thermal insulator, not a conductor, so don't overdo it at this point.

WHERE THE FAN PLUGS IN. The cooler's fan goes on the four pin header marked CPU_FAN, right beside the socket. It is the one header the firmware watches, and a fan anywhere else leaves it reading zero rpm on the processor, which most boards refuse to boot on.

Memory or RAM

A G.SKILL RIPJAWS S5 KIT, 64 GB IN TWO 32 GB MODULES, part number F5-6000J3636F32GX2-RS5K⁵:

- RAM keeps a weight file warm between loads, plus room for the containers layered on later.
- For speed the kit runs DDR5-6000, double data rate, at CL36-36-36-96 column address strobe latency and 1.35 volt. Its AMD EXPO profile, extended profiles for overclocking, is where we will take this from in the configuration later on.
- We stay at two modules as filling all four slots pushes the memory controller down a speed grade.



INSTALL IT by seating one module in the second slot of each channel, follow the marks on the motherboard or its manual, until the clips snap up (Figure 4).

THE FIVE FAN HEADERS. Each is four pin and good for 24 watt.

- CPU_FAN: the cooler's fan goes here.
- CPU_OPT: a second CPU fan or pump, right next to it, useful if your tower cooler is a dual fan model.
- SYS_FAN1, SYS_FAN2, SYS_FAN3_PUMP: the three for the fans you zip tie to the frame rail later.

⁵ G.SKILL. Ripjaws S5 DDR5-6000 CL36 64GB (2x32GB), F5-6000J3636F32GX2-RS5K. Product datasheet, 2026

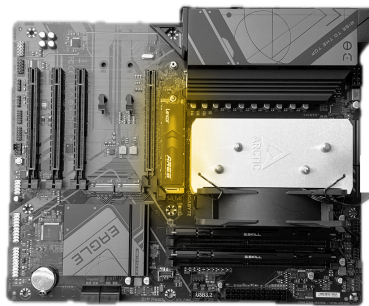
Figure 4: G.SKILL Ripjaws S5 64 GB DDR5-6000

WHY 64 AND NOT 128. We will be doing fine with 64GB. The one real case for a large kit is offloading a model too big for total VRAM into system memory, but then again you could just add another 3090 and have 48GB of VRAM to play with.

Storage

ONE NVME DRIVE, for non volatile memory express, 2 TB in the M.2 2280 size and wired PCIe 4.0 x4, holding both the system and the model library ⁶:

- For speed it belongs in the first M.2 socket, the only one wired straight from the processor. A quantised model is tens of gigabytes read start to finish every time it loads, and that socket is what sets how long we wait on a model swap.
- For width the socket is PCIe 5.0 x4 while our drive is Gen4 x4, so it runs at its own full width, around 7.4 GB/s, which is already fast but also leaves headroom left in the socket for a faster drive later.
- We go for single storage as the second socket runs only PCIe 4.0 x2 and would halve this drive's lanes, and the third sits behind the chipset, sharing its uplink with network and USB.



⁶ GIGABYTE. B650 EAGLE AX (rev. 1.2) motherboard specifications. Product datasheet, 2026

Figure 5: NVMe Gen4 x4 2TB, M.2 2280

FIT THE DRIVE in the first socket, in at an angle, pressed flat, and held down with its single screw or latch (Figure 5).

THE TERABYTES ARE A USE-CASE CALL. It leaves room for a few dozen quantised models beside the system, which is plenty for pure inference on a handful of models. If you need more models or upgrade regularly pruning will become a housekeeping chore early one, which is best practice anyway in my opinion. Datasets, fine tuning runs and their checkpoints on the other hand quickly will become an issue with 2TB, so size the drive to what you actually intend to do.

The Frame

AN OPEN GPU RACK FRAME. Any aluminium skeleton will do, a flat bed for the board and a rail above it that holds cards upright.

- For the card, an open bed means it hangs in room air rather than behind a panel, so it runs cooler than the same card in a mid tower.
- For growth, the rail makes it easy to swap and add cards easily.
- Of course this also gives us a server style look, which I appreciate in this project.

CASES ARE EXPENSIVE. The downside is that the rig is open to dust and to hands, so if you can not move it out of the way - reconsider a case. Make sure it holds your motherboard and parts in any case. Yep, pun intended.

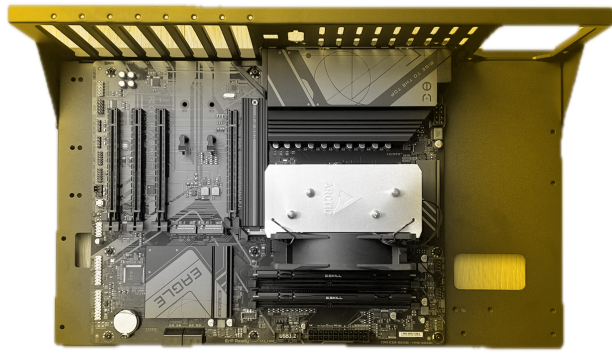


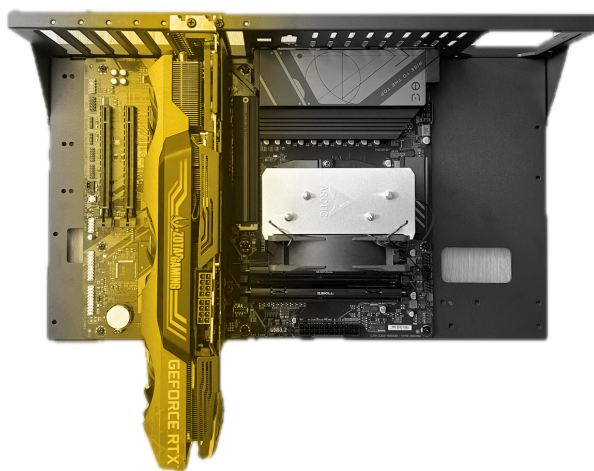
Figure 6: Open GPU rack frame

MOUNT IT NOW. Assemble the frame, screw the standoffs into the bed, and lower the populated board onto them. I directly screwed this whole thing into a wall, but you can also put it on a shelf - if you prefer to not go rogue in your flat. The frame's two pin switch is now wired onto PW on F_PANEL.

The Graphics Card

A RTX 3090, 24 GB seated straight in the board's top slot:

- For capacity, 24 GB of GDDR6X VRAM on a 384 bit bus, which is the limiting factor for model size and model context.
- For token speed, the 936 GB/s of memory bandwidth⁷ matters more than the 10 496 CUDA cores while picking a GPU.
- For the link, the top slot is the one PCIe 4.0 x16 connection in this machine, as long as you go with a single card.



WHY THE 3090 and not:

- A6000, 48 GB: Twice the VRAM at ten times the price.
- 5090, 32 GB: More bandwidth, but no NVLink.
- 4090, 24 GB: The same VRAM at twice the price.
- Mac or DGX Spark: Unified memory crawls on a dense 70 billion parameter model.
- 3090 Founders Edition: wants NVIDIA's twelve pin adapter rather than plain eight pin cables.

⁷ NVIDIA. GeForce RTX 3090 family specifications. Product datasheet, 2022

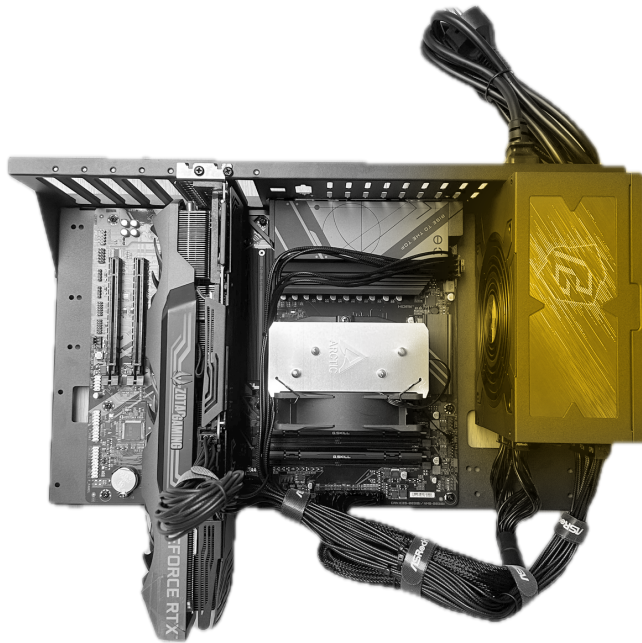
Figure 7: RTX 3090 24 GB

FIT THE CARD into the top slot, pressed down evenly until the retention clip snaps shut, fasten its bracket, and leave its power cables until the supply is mounted. The multi-card rig extensions this motherboard is able to hold will be detailed out in a later section. This in itself is a powerful machine.

The Power Supply

AN ASROCK PHANTOM GAMING PG-1600G, 1600 watt against the roughly 500 watt this machine draws, bought for the headroom⁸:

- For the graphics card, a 3090 averages 350 watt and spikes far above it, which is what trips undersized supplies mid run. ATX 3.1 rates this unit for 220 percent of that, and the 1600 watt rating is what we need for four potential cards.
- For the board, one 24 pin main cable and one processor cable, 4+4 pin, both modular, so only what we use leaves the unit.
- Eight PCIe cables ship in the box. We wire two, and the rest is already there for later cards.



WIRE IT LAST. Mount the supply on the frame, then run the cables in this order:

BUYING USED. Buy only from a seller who shows a test of the card under load, and budget a thermal pad replacement. For the used market, a three fan model with two eight pin connectors, an EVGA FTW3 Ultra, MSI Suprim or Gaming, Gigabyte Gaming OC, or ASUS TUF.

⁸ ASRock. Phantom Gaming PG-1600G 1600 W ATX 3.1 power supply. Product datasheet, 2026

GET SOLAR POWER FOR YOUR IDLE WATTAGE. The floor runs day and night whether or not anyone prompts the machine, and most of it is the platform rather than the card. A steady draw is what a balcony solar setup covers best: panels and a capped inverter suit a constant load, not the spikes this supply was bought to survive. Put a meter on the wall socket before sizing any of it.

Figure 8: ASRock PG-1600G, 1600 W

WHAT STAYS IN THE BOX. The unit ships two processor cables and this board takes one, so there is no second socket to hunt for. The two 12V-2x6 cables, 600 watt each, are for 40 and 50 series cards, while a 3090 takes plain eight pin. Twelve SATA and six Molex wait for spinning storage or a powered riser.

1. The 24 pin main cable (often it is labelled MB) into ATX, the widest socket on the board, sitting on the long edge alongside the memory slots. Press until the latch clicks.
2. The 4+4 pin processor cable, from the socket marked CPU on the supply into ATX_12V on the board, on the short edge past the cooler, beside the VRM heatsink nearest the rear I/O panel.
3. The two PCIe cables to the card's own two eight pin connectors, along its top edge, one cable each and never daisy chained.

LEAVE THE MAINS LEAD UNPLUGGED. The rig is now fully assembled, wired, and ready to be powered on for the first time. Skip to [the next chapter](#) for the BIOS setup and the first boot of the rig.

Multi-Card Rig

ONE TO THREE MORE RTX 3090s find a place on this rig design⁹. A second card buys memory for larger models, while the third and fourth cards buy additional throughput.

⁹ GIGABYTE. B650 EAGLE AX (rev. 1.2) motherboard specifications. Product datasheet, 2026

Card Two

STAY ON THE 3090. It is the only GeForce 30 card with NVLink, so a matched pair takes a bridge and pools to 48 GB for one model. Mixed cards never pool.

WHAT NVLINK IS. A bridge that clips onto the top edge of two cards and carries 112.5 GB/s bidirectional traffic on the 3090.

THE REST IS MECHANICAL. Drop the card into the next x16 sized position, wired PCIe 3.0 x1, on a plain riser. Riser spacing is yours to set, so match it to the bridge you buy, get it fixed on your frame, and run its own two cables from the supply.

Cards Three and Four

ADD THESE WITH INCREASED WORKLOAD. Two x16 sized positions remain, one from the processor and one from the chipset. Both are wired PCIe 3.0 x1: full length connector, a single lane behind it, about 1 GB/s only. Keep it at one model per (linked) card(s). A model split across cards parallelism moves activations every token and starts to crawl. So the idea is to have the same model loaded on each card, and let the scheduler round robin between them for throughput.

4 CARDS WILL DEFINITELY GET YOU FANS. Zip tie a bank of case fans along the rail, undervolt before the fourth card goes in.

ON POWER. Four cards at 350 watt plus the platform reach roughly 1550 watt, which becomes uncomfortable against our supply's 1600 limit. Undervolted near 0.875 volt each card draws about 280 and the whole rig lands near 1270, which makes it comfortable again.

The Minimal Viable Rig

A RASPBERRY PI 5 WITH 8 GB, (go 16 GB if you can) an active cooler, the official 5 volt 5 amp USB-C supply and a 64 GB microSD card¹⁰, will enable you to learn just as much on a tighter budget.

THE 8 GB IS SHARED: processor and graphics core read the same LPDDR4X, so it is system and video memory at once. A four billion parameter model at four bit quantisation takes near 2.5 GB and leaves five for the operating system and containers. What it costs mainly is speed as LPDDR4X-4267 moves roughly 17 GB/s where a 3090 moves 936, and generation reads the whole model once per token, so answers arrive at single digit tokens per second rather than dozens. But hey we don't care about speed for learning, and buying hardware later on is the easy part.

BUILD IT IN FIVE MINUTES. Clip the active cooler onto the board and plug its fan into the header beside it, write the operating system image onto the microSD card and seat the card in the slot on the underside, and leave the USB-C supply out of the wall. A complete machine, wired and never switched on, in one hand.

THE METAL IS FINISHED, whichever of the two machines is on your bench. What it still lacks is a first power on and a handful of BIOS values that make a card addressable, and then an operating system, a driver, and weights loaded into memory together with some orchestration. Let's get on it.

¹⁰ Raspberry Pi Ltd. Raspberry Pi 5 product brief. Product datasheet, 2023

THE ON-A-BUDGET-THREAD. Same operating system, same container runtime, same Ollama binary on the same port, same clients pointed at it. Only the model and performance shrinks.

The Stack

2026-08-20 · *daring lingonberry Lachs*

SILICON, MEET SOFTWARE.

THIS CHAPTER INSTALLS ONE STACK: Ubuntu Server on the rig you just wired, the NVIDIA driver under it, and Ollama on top, which is enough to answer an API call on weights held in your own video memory. Nothing above the metal changes when a second card arrives. Install the layers in Table 2 in the order of the sections that follow, and run each layer’s check before starting the next, until the rig answers on the port and reports on itself. Day one is one evening, most of it spent waiting on downloads.

THE STACK IS EIGHT LAYERS and each one is a single command with a single answer. Nothing in it is compiled from source, and only the last layer takes a configuration file.

Layer	Piece	Proves it
BIOS	five values in setup	re-enter setup, all five hold
OS	Ubuntu Server 26.04 LTS	<code>lspci grep -i nvidia</code>
GPU driver	NVIDIA proprietary driver	<code>nvidia-smi</code>
Containers	Docker Engine	<code>docker run hello-world</code>
Passthrough	NVIDIA Container Toolkit	<code>docker run --gpus all</code>
Inference server	Ollama	<code>curl localhost:11434</code>
Model weights	Qwen3.8 27B at four bits	<code>ollama run qwen3.8:27b</code>
Monitoring	Netdata and a heartbeat	<code>curl -sI localhost:19999</code>

Table 2: The stack, in the order it goes on. The third column is the whole troubleshooting method: run it before starting the next layer, and a failure is located rather than guessed at.

The BIOS

FIVE VALUES IN THE BIOS SETUP UTILITY, three of which decide whether a graphics card is addressable at all¹¹:

- For addressing, Above 4G Decoding maps a card's base address register windows past the 32 bit boundary, without which a second card never maps.
- For loading, Resizable BAR hands the processor a card's whole 24 GB in one window instead of a 256 MB sliding one.
- For passthrough, IOMMU, the input output memory management unit, maps card memory into a container.
- For VRAM, Initial Display Output on IGD keeps the monitor on the integrated graphics and all 24 GB free for models.
- For memory speed, the EXPO profile clocks the kit at its rated 6000 rather than the conservative JEDEC default a DDR5 kit boots at.

SET THEM NOW. Monitor on the board's own display output, not on the card. Switch the supply on and press the power button; fans that turn half a revolution and stop mean the 24 pin is not seated. Press Del during POST, the power on self test, then save and exit:

```
Above 4G Decoding ..... Enabled
Re-Size BAR Support ..... Enabled
IOMMU ..... Enabled
Initial Display Output ..... IGD
EXPO / XMP profile ..... Profile 1
```

The OS

UBUNTU SERVER 26.04 LTS, written to a USB stick with the same imager you would use for any machine, with no desktop environment on it¹²:

- For the driver, Canonical ships the proprietary NVIDIA driver in the APT archive, so a supported card costs one command rather than a vendor runfile and a blacklisted module.
- For the card, a server install runs no display server, neither X nor Wayland, so nothing on the box holds VRAM or wakes a screen.
- For the library, manual partitioning in the installer leaves the rest of the drive as a separate filesystem for weights.

¹¹ GIGABYTE. B650 EAGLE AX (rev. 1.2) motherboard specifications. Product datasheet, 2026

MENU NAMES MOVE. BIOS here is the UEFI firmware that Gigabyte still labels BIOS, and these five sit under different headings in different revisions of it, so use the setup search rather than a path from a screenshot. Set all five in one visit: a value missed here surfaces later as a hardware fault that is not one.

¹² Canonical. Ubuntu server documentation: Basic installation. Ubuntu Server Documentation, 2026b

WHERE THE WEIGHTS LIVE. Point OLLAMA.MODELS at that second filesystem once Ollama is installed, and reinstalling the operating system never touches a library that took days to download.

WORK OVER SSH. Enable OpenSSH in the installer, then unplug the monitor and do everything that follows from your laptop. The first two commands on the fresh install name the system and the card, which is the last look at the machine before a driver can be blamed for anything:

```
$ lsb_release -a          # Ubuntu 26.04 LTS
$ lspci | grep -i nvidia   # NVIDIA GA102 [RTX 3090]
```

The GPU Driver

THE PROPRIETARY NVIDIA DRIVER from the Ubuntu archive, a kernel module and the userspace libraries that make the card visible to every program above it¹³:

- For selection, `ubuntu-drivers` reads the card's PCI device id and picks the branch that supports it, so there is no version to choose.
- For CUDA, the package provides `libcuda.so`, the driver API and the only CUDA library Ollama loads from the system, so there is no CUDA Toolkit to install for inference.
- For proof, `nvidia-smi` prints the card, its 24 GiB, and the driver version. Until it does, nothing above this layer can work.

INSTALL IT AND REBOOT, because a running kernel keeps the module it already loaded:

```
$ sudo ubuntu-drivers install
$ sudo reboot
$ nvidia-smi          # the 3090, with its 24 GiB
```

The Container Runtime

DOCKER ENGINE AND THE NVIDIA CONTAINER TOOLKIT, one pair, because a container runtime on its own cannot reach a graphics card¹⁴:

- For the card, the toolkit installs an OCI runtime hook that mounts the host driver into a container at start, so it is installed once on the metal and never inside an image.
- For upgrades, every service above this layer ships as an image, so an upgrade is a pulled tag and a rollback is the previous one.
- For proof, `--gpu all` makes `nvidia-smi` print from inside a container the same card it printed on the host.

NO DRIVER IN THAT LINE. `lspci` reads what the BIOS put on the PCIe bus, so a card listed there is seated, powered, and addressable whatever happens above it. A card missing there is a hardware fault, and no amount of installing fixes it. The driver and the container runtime go onto this install in the same sitting.
¹³ Canonical. NVIDIA drivers installation. Ubuntu Server Documentation, 2026a

¹⁴ NVIDIA. Installing the NVIDIA container toolkit. NVIDIA Container Toolkit Documentation, 2025

WHY CONTAINERS AT ALL, when Ollama itself runs on the metal. Everything layered on later asks Docker for the GPU and gets it, and several such services share one card without any of them carrying a driver of its own.

WIRE THEM TOGETHER. `nvidia-ctk` writes the runtime into Docker's `daemon.json`, which takes a restart before it applies¹⁵:

```
$ curl -fsSL https://get.docker.com | sh
$ sudo nvidia-ctk runtime configure --runtime=docker
$ sudo systemctl restart docker
$ docker run --rm --gpus all ubuntu nvidia-smi
```

¹⁵ Docker. Install Docker Engine on Ubuntu. Docker Documentation, 2026

The Inference Server

OLLAMA IS THE INFERENCE SERVER, the process that loads weights into video memory, holds them there between requests, and answers over HTTP¹⁶:

- For setup, the install script registers `ollama.service` under `systemd`, and a model that refuses to load says why in `journalctl -u ollama`.
- For clients, it serves an OpenAI-compatible API at `:11434/v1`, the shape every chat front end, editor, and agent speaks.
- For the library, weights land wherever `OLLAMA_MODELS` points, one directory of content addressed blobs shared across every tag.
- For reach, it binds to localhost and reads no authorisation header, so publishing that port without a gateway hands the card to whoever finds it.

INSTALL IT AND POINT THE LIBRARY at the second filesystem, in a `systemd` drop in override rather than in the shipped unit file:

```
$ curl -fsSL https://ollama.com/install.sh | sh
$ sudo systemctl edit ollama
[Service]
    Environment="OLLAMA_MODELS=/srv/models"
$ sudo systemctl restart ollama
$ curl localhost:11434          # Ollama is running
```

¹⁶ Ollama. Ollama documentation: Linux install and OpenAI-compatible API. Ollama Documentation, 2026a

SERVER, NOT ENGINE. The arithmetic runs in GGML and the CUDA kernels it ships, which is the inference engine. Ollama wraps it with the three things kernels have no opinion about: which weights stay resident, whose request runs next, and what the API looks like. Swapping the server replaces all three and leaves the card and the driver alone.

The Model Weights

THE MODEL IS ONE FILE of a few gigabytes that Ollama maps into video memory and reads once per token, so a change of mind is a download¹⁷:

- For the budget, 24 GB is all of it and quantization decides what fits: near two thirds of a gigabyte per billion parameters at the four bit tags that ship by default, so twenty seven billion cost 18 GB.
- For the context, the window is a KV cache allocated on top of the weights, so a 20 GB model with a long window tips the card into system memory and crawls.
- For the work, tool calling is a filter and not a preference, since a model that cannot call a tool cannot flip a switch or search a note.

FIVE THAT FIT the card, with 18 GB the working default and roughly six left for the KV cache and a small embedding model.

Tag	Weights	Picked for
qwen3.5:9b	6.6 GB	quick, 256K window
gpt-oss:20b	14 GB	reasoning and agents
mistral-small3.2:24b	15 GB	instruction following
qwen3.8:27b	18 GB	the default here
gemma4:31b	20 GB	the most that fits

PULL ONE AND PROVE IT ON THE API, the only interface that has to work:

```
$ ollama pull qwen3.8:27b
$ curl localhost:11434/v1/chat/completions \
  -H "Authorization: Bearer ollama" \
  -d '{"model": "qwen3.8:27b", "messages":
    [{"role": "user", "content": "say hello"}]}'
```

Monitoring

NETDATA DRAWS THE RIG, every core, disk, container, and the card itself, at a sample a second, from one install script and with no dashboard to build¹⁸:

- For the card, one collector reads `nvidia-smi` and charts temperature, power draw, framebuffer memory, and utilisation.

¹⁷ Ollama. Model library: Tags, quantization, and context windows. Ollama Model Library, 2026b

THESE TAGS WILL AGE. Qwen went from 3 to 3.5 to 3.6 to 3.8 inside a year, while the budget does not move: take the tag that costs the same gigabytes as the one you run now. The same weights are 30 GB at eight bits and 56 GB at full precision, which is what puts them out of reach. The bearer string below is a placeholder, since Ollama reads no authorisation header, but OpenAI-compatible clients refuse to send a request without one.

¹⁸ Netdata. Netdata agent: Installation, the Nvidia GPU collector, and alert notifications. Netdata Documentation, 2026

- For history, per second samples roll into per minute and per hour tiers, so a month of usage sits on disk with no database beside it.
- For reach, the dashboard is a page on :19999 with no account and no cloud behind it, served by the machine it measures.

INSTALL IT AND SWITCH THE CARD ON. The GPU collector ships disabled, and it is why monitoring goes on the metal rather than into Docker: it shells out to `nvidia-smi` on the host.

```
$ curl https://get.netdata.cloud/kickstart.sh \
  > /tmp/kickstart.sh && sh /tmp/kickstart.sh
```

```
$ cd /etc/netdata
$ sudo ./edit-config go.d/nvidia_smi.conf
  jobs:
    - name: nvidia_smi
```

```
$ sudo systemctl restart netdata
$ curl -sI localhost:19999    # 200 OK
```

Alerting and Metering

A DISCORD WEBHOOK is a URL that turns an HTTP POST into a message in a channel, which makes it the cheapest alerting target a household can own¹⁹:

- For the alarms, Netdata posts its own thresholds to it, so a hot card, a full disk, or a stopped container reaches your phone with no rule to write.
- For the silence, an alarm on the machine it watches reports every failure except the one that matters, because a dead machine sends nothing. A heartbeat inverts that: the message that fails to arrive is the alert.
- For the secret, the webhook URL is the credential, so it lives in a file that only root can read.

HAND THE WEBHOOK TO NETDATA, created in the Discord channel settings under Integrations:

```
$ sudo ./edit-config health_alarm_notify.conf
  SEND_DISCORD="YES"
  DISCORD_WEBHOOK_URL="https://discord.com/api/..."
  DEFAULT_RECIPIENT_DISCORD="alerts"
```

READ THREE CHARTS FIRST. A card past 83 degrees is protecting itself and losing clock, framebuffer memory pinned at 24 GB with an answer still crawling means the model spilled into system memory, and the filesystem holding the library filling up is the failure that arrives quietly and stops every pull at once.

¹⁹ Discord. Intro to webhooks. Discord Developer Documentation, 2026

DAILY, NOT HOURLY. A channel that pings every hour stops being read inside a week, and an alert nobody reads is not an alert. One message a morning is enough for a household to notice a missing one by lunchtime. Let `jq` build the JSON body too: a model tag carrying a quote produces a request Discord rejects with no useful reason.

ADD THE HEARTBEAT, which names the resident model once a morning. It runs from the standard systemd pair, a oneshot service and a timer with `OnCalendar=daily` and `Persistent=true`, so a reboot delays the message rather than swallowing it²⁰:

```
# /usr/local/bin/rig-heartbeat
s="rig up | $(curl -sf localhost:11434/api/ps \
  | jq -r '.models[0].name // "idle"')"\
curl -s -H 'Content-Type: application/json' \
  -d "$(jq -nc --arg c "$s" '{content:$c}')" \
  "$(cat /etc/rig-webhook)"
```

²⁰ systemd. systemd.timer: Timer unit configuration. systemd Manual Pages, 2026

METER IT WITH A SECOND TIMER at `OnCalendar=minutely`, appending the month, the resident model, and the card's watts to `/var/log/rig-usage`. One awk pass on the first of the month turns that file into hours per model and kilowatt hours, posted to the same webhook:

```
2026-07
qwen3.8:27b 61.4 h
qwen3.5:9b 8.9 h
idle 673.7 h
33.16 kWh
```

Multi-Card Serving

OLLAMA SERVES A HOUSEHOLD FROM ONE CARD and keeps serving when the second, third, and fourth arrive. What changes is the shape of the load: several people prompting at once, or an agent firing dozens of requests a minute, is throughput rather than conversation, and throughput is a different inference server.

VLLM WINS ON CONCURRENCY. It runs continuous batching over a paged KV cache instead of queueing requests, so the card keeps working while one prompt waits on another²¹. Both servers speak the same OpenAI-compatible API, so the swap is a different container on the same port and every client keeps working untouched.

KEEP ONE MODEL PER CARD. Cards two to four sit on one PCIe lane, roughly 1 GB/s. Tensor parallelism, which splits one model across cards, pushes activations over that link on every token and crawls, while pipeline parallelism and a plain replica per card send almost nothing over it. Let the scheduler round robin between replicas:

WHAT THE LEDGER MEASURES is resident minutes, not tokens. Ollama holds a model in video memory for five minutes past the last request, so a month of short sessions reads high. What it answers honestly is which model earned its place on the card and what the month drew from the wall: two thirds of those 33 kilowatt hours are idle at roughly 20 watts.

²¹ vLLM Project. Distributed inference and serving. vLLM Documentation, 2025

NOT YET, ON ONE CARD. Serving a household from a single 3090, one install command beats a tuning session, and the queue never runs deep enough for batching to pay for itself. Swap when people start waiting on each other, not before.

throughput rises with the card count, a single answer arrives no faster, and total video memory still sets the largest model that will run.

The Minimal Viable Rig

A RASPBERRY PI 5 RUNS THIS CHAPTER without the BIOS section and without the GPU driver, because there is no setup utility to visit and no card to make visible²²:

- For the system, Ubuntu Server 26.04 for arm64 writes to the microSD from the same imager and boots with OpenSSH the same way.
- For the runtime, the Docker install script and the Ollama install script both detect the architecture and need no flag for it.
- For the model, four billion parameters at four bit quantisation take near 3.4 GB of the shared 8 GB and leave four for the system and containers.

PULL THE SMALLER MODEL. Substitute `ollama pull qwen3.5:4b` for the twenty seven billion pull and every command downstream is unchanged, same port, same API, same clients. What it costs is speed: LPDDR4X-4267 moves roughly 17 GB/s against the 3090's 936, and generation reads the whole model once per token, so answers arrive at single digit tokens per second rather than dozens.

²² Raspberry Pi Ltd. Raspberry Pi 5 product brief. Product datasheet, 2023

EIGHT GIGABYTES IS THE FLOOR. The 16 GB board is the headroom upgrade and it buys a larger model rather than a faster one, since the memory bandwidth is the same either way. There is no `lspci` line to look for and no `nvidia-smi`: the 8 GB is system and video memory at once.

WHAT YOU NOW OWN is a served rig: one request returns a completion in the shape every client speaks, on weights in your own video memory, with nothing metered and no prompt leaving the box, and a message every morning proving the machine is still up.

OLLAMA RUNS, BUT ONLY ON THE BOX ITSELF. It listens on localhost, checks no password, and the key in that last request was a string it never read. The next chapter carries it to your phone, your laptop, and your terminal wherever they are, and gives it a credential that means something, without opening the card to everyone.

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